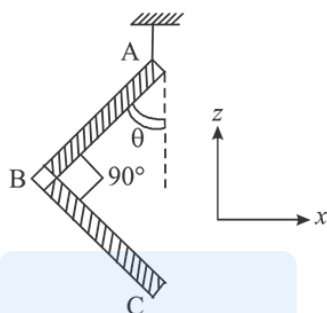


JEE Mains 2019 Chapter wise Question Bank

Center of Mass, Momentum Conservation and Collision - Questions

Q1

An L-shaped object, made of thin rods of uniform mass density, is suspended with a string as shown in figure. If $AB = BC$, and the angle made by AB with downward vertical is θ , then:



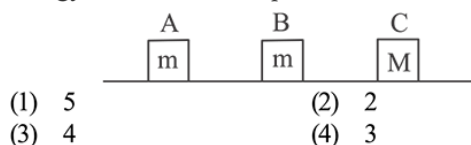
- (1) $\tan \theta = \frac{1}{2\sqrt{3}}$ (2) $\tan \theta = \frac{1}{2}$
 (3) $\tan \theta = \frac{2}{\sqrt{3}}$ (4) $\tan \theta = \frac{1}{3}$

9 Jan Morning

Q2

Three blocks A, B and C are lying on a smooth horizontal surface, as shown in the figure. A and B have equal masses, m while C has mass M . Block A is given an initial speed v towards B due to which it collides with B perfectly inelastically. The combined mass collides with C,

also perfectly inelastically $\frac{5}{6}$ th of the initial kinetic energy is lost in whole process. What is value of M/m ?



9 Jan Morning

Q3

A piece of wood of mass 0.03 kg is dropped from the top of a 100 m height building. At the same time, a bullet of mass 0.02 kg is fired vertically upward, with a velocity 100 ms^{-1} , from the ground. The bullet gets embedded in the wood. Then the maximum height to which the combined system reaches above the top of the building before falling below is: ($g = 10 \text{ ms}^{-2}$)

- (1) 20 m (2) 30 m (3) 40 m (4) 10 m

10 Jan Morning

Q4

A liquid of density ρ is coming out of a hose pipe of radius a with horizontal speed v and hits a mesh. 50% of the liquid passes through the mesh unaffected. 25% loses all of its momentum and 25% comes back with the same speed. The resultant pressure on the mesh will be:

- (1) $\frac{1}{4}\rho v^2$ (2) $\frac{3}{4}\rho v^2$
 (3) $\frac{1}{2}\rho v^2$ (4) ρv^2

11 Jan Morning

Q5

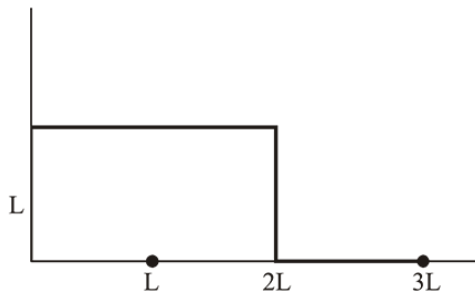
A simple pendulum, made of a string of length l and a bob of mass m , is released from a small angle θ_0 . It strikes a block of mass M , kept on a horizontal surface at its lowest point of oscillations, elastically. It bounces back and goes up to an angle θ_1 . The M is given by:

- (1) $\frac{m}{2} \left(\frac{\theta_0 + \theta_1}{\theta_0 - \theta_1} \right)$ (2) $m \left(\frac{\theta_0 - \theta_1}{\theta_0 + \theta_1} \right)$
 (3) $m \left(\frac{\theta_0 + \theta_1}{\theta_0 - \theta_1} \right)$ (4) $\frac{m}{2} \left(\frac{\theta_0 - \theta_1}{\theta_0 + \theta_1} \right)$

12 Jan Morning

Q6

The position vector of the centre of mass r_{cm} of an asymmetric uniform bar of negligible area of cross-section as shown in figure is:



(1) $r_{cm} = \frac{13}{8}L\hat{x} + \frac{5}{8}L\hat{y}$ (2) $r_{cm} = \frac{5}{8}L\hat{x} + \frac{13}{8}L\hat{y}$

(3) $r_{cm} = \frac{3}{8}L\hat{x} + \frac{11}{8}L\hat{y}$ (4) $r_{cm} = \frac{11}{8}L\hat{x} + \frac{3}{8}L\hat{y}$

12 Jan Morning

Q7

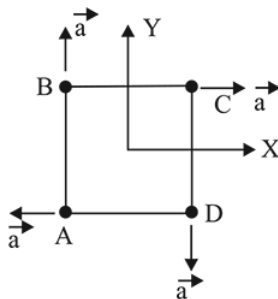
An alpha-particle of mass m suffers 1-dimensional elastic collision with a nucleus at rest of unknown mass. It is scattered directly backwards losing 64% of its initial kinetic energy. The mass of the nucleus is :

- (1) $2m$ (2) $3.5m$
(3) $1.5m$ (4) $4m$

12 Jan Evening

Q8

Four particles A, B, C and D with masses $m_A = m$, $m_B = 2m$, $m_C = 3m$ and $m_D = 4m$ are at the corners of a square. They have accelerations of equal magnitude with directions as shown. The acceleration of the centre of mass of the particles is :



- (1) $\frac{a}{5}(\hat{i} - \hat{j})$ (2) $a(\hat{i} + \hat{j})$ (3) Zero (4) $\frac{a}{5}(\hat{i} + \hat{j})$

8 April Morning

Q9

A body of mass m_1 moving with an unknown velocity of $v_1\hat{i}$, undergoes a collinear collision with a body of mass m_2 moving with a velocity $v_2\hat{i}$. After collision, m_1 and m_2 move with velocities of $v_3\hat{i}$ and $v_4\hat{i}$, respectively.

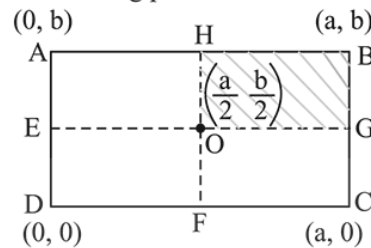
If $m_2 = 0.5 m_1$ and $v_3 = 0.5 v_1$, then v_1 is :

- (1) $v_4 - \frac{v_2}{2}$ (2) $v_4 - v_2$ (3) $v_4 - \frac{v_2}{4}$ (4) $v_4 + v_2$

8 April Evening

Q10

A uniform rectangular thin sheet ABCD of mass M has length a and breadth b , as shown in the figure. If the shaded portion HBGO is cut-off, the coordinates of the centre of mass of the remaining portion will be :



- (1) $\left(\frac{3a}{4}, \frac{3b}{4}\right)$ (2) $\left(\frac{5a}{3}, \frac{5b}{3}\right)$
(3) $\left(\frac{2a}{3}, \frac{2b}{3}\right)$ (4) $\left(\frac{5a}{12}, \frac{5b}{12}\right)$

8 April Evening

Q11

A body of mass 2 kg makes an elastic collision with a second body at rest and continues to move in the original direction but with one fourth of its original speed. What is the mass of the second body?

- (1) 1.0 kg (2) 1.5 kg
(3) 1.8 kg (4) 1.2 kg

9 April Morning

Q12

A particle of mass ' m ' is moving with speed ' $2v$ ' and collides with a mass ' $2m$ ' moving with speed ' v ' in the same direction. After collision, the first mass is stopped completely while the second one splits into two particles each of mass ' m ', which move at angle 45° with respect to the original direction.

The speed of each of the moving particle will be:

- (1) $\sqrt{2} v$ (2) $2\sqrt{2} v$
 (3) $v/(2\sqrt{2})$ (4) $v/\sqrt{2}$

9 April Evening

Q13

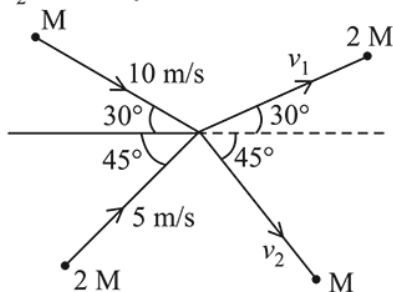
A wedge of mass $M = 4m$ lies on a frictionless plane. A particle of mass m approaches the wedge with speed v . There is no friction between the particle and the plane or between the particle and the wedge. The maximum height climbed by the particle on the wedge is given by:

- (1) $\frac{v^2}{g}$ (2) $\frac{2v^2}{7g}$ (3) $\frac{2v^2}{5g}$ (4) $\frac{v^2}{2g}$

9 April Evening

Q14

Two particles, of masses M and $2M$, moving, as shown, with speeds of 10 m/s and 5 m/s , collide elastically at the origin. After the collision, they move along the indicated directions with speeds v_1 and v_2 , respectively. The values of v_1 and v_2 are nearly :



- (1) 6.5 m/s and 6.3 m/s (2) 3.2 m/s and 6.3 m/s
 (3) 6.5 m/s and 3.2 m/s (4) 3.2 m/s and 12.6 m/s

10 April Morning

Q15

When M_1 gram of ice at -10°C (Specific heat = $0.5 \text{ cal g}^{-1}^\circ\text{C}^{-1}$) is added to M_2 gram of water at 50°C , finally no ice is left and the water is at 0°C . The value of latent heat of ice, in cal g^{-1} is:

- (1) $\frac{50M_2}{M_1} - 5$ (2) $\frac{5M_1}{M_2} - 50$
 (3) $\frac{50M_2}{M_1}$ (4) $\frac{5M_2}{M_1} - 5$

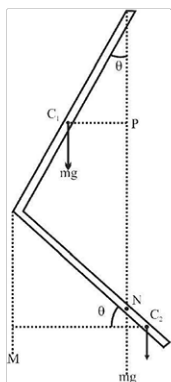
12 April Evening

JEE Mains 2019 Chapter wise Question Bank

Center of Mass, Momentum Conservation and Collision - Answers

Q1

(4)



Given that, the rod is of uniform mass density and $AB = BC$

Let mass of one rod is m .

Balancing torque about hinge point.

$mg(C_1P) = mg(C_2N)$

$$mg\left(\frac{L}{2}\sin\theta\right) = mg\left(\frac{L}{2}\cos\theta - L\sin\theta\right)$$

$$\Rightarrow \frac{3}{2}mgL\sin\theta = \frac{mgL}{2}\cos\theta \Rightarrow \frac{\sin\theta}{\cos\theta} = \frac{1}{3} \text{ or, } \tan\theta = \frac{1}{3}$$

9 Jan Morning

Q2

(3) Kinetic energy of block A

$$k_1 = \frac{1}{2}mv_0^2$$

$\therefore$ From principle of linear momentum conservation

$$mv_0 = (2m + M)v_f \Rightarrow v_f = \frac{mv_0}{2m + M}$$

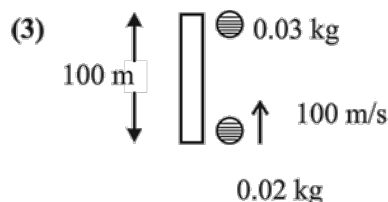
According to question, of $\frac{5}{6}$ th the initial kinetic energy is lost in whole process.

$$\therefore \frac{k_i}{k_f} = 6 \Rightarrow \frac{\frac{1}{2}mv_0^2}{\frac{1}{2}(2m + M)\left(\frac{mv_0}{2m + M}\right)^2} = 6$$

$$\Rightarrow \frac{2m + M}{m} = 6 \therefore \frac{M}{m} = 4$$

9 Jan Morning

Q3



Time taken for the particles to collide,

$$t = \frac{d}{V_{rel}} = \frac{100}{100} = 1 \text{ sec}$$

Speed of wood just before collision $= gt = 10 \text{ m/s}$

and speed of bullet just before collision $= v - gt$
 $= 100 - 10 = 90 \text{ m/s}$

$$S = 100 \times 1 - \frac{1}{2} \times 10 \times 1 = 95 \text{ m}$$

Now, using conservation of linear momentum just before and after the collision

$$-(0.03)(10) + (0.02)(90) = (0.05)v$$

$$\Rightarrow 150 = 5v$$

$$\therefore v = 30 \text{ m/s}$$

Max. height reached by body

$$h = \frac{v^2}{2g} = \frac{30 \times 30}{2 \times 10} = 45 \text{ m}$$

Before	After
0.03 kg 10 m/s	0.05 kg
0.02 kg 90 m/s	

$\therefore$ Height above tower $= 40 \text{ m}$

10 Jan Morning

Q4

(2) Mass per unit time of the liquid = ρav
Momentum per second carried by liquid
= $\rho av \times v$

Net force due to bounced back liquid,

$$F_1 = 2 \times \left[\frac{1}{4} \rho av^2 \right]$$

Net force due to stopped liquid, $F_2 = \frac{1}{4} \rho av^2$

Total force,

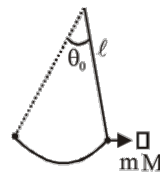
$$F = F_1 + F_2 = \frac{1}{2} \rho av^2 + \frac{1}{4} \rho av^2 = \frac{3}{4} \rho av^2$$

Net pressure = $\frac{3}{4} \rho v^2$

11 Jan Morning

Q5

(2)



Velocity before collision $v = \sqrt{2g\ell(1 - \cos\theta_0)}$

Velocity after collision

$$v_1 = \sqrt{2g\ell(1 - \cos\theta_1)}$$

Using momentum conservation

$$mv = MV_m - mV_1$$

$$m\sqrt{2g\ell(1 - \cos\theta_0)} = MV_m - m\sqrt{2g\ell(1 - \cos\theta_1)}$$

$$\Rightarrow m\sqrt{2g\ell} \{ \sqrt{1 - \cos\theta_0} + \sqrt{1 - \cos\theta_1} \} = MV_m$$

$$\text{and } e = 1 = \frac{V_m + \sqrt{2g\ell(1 - \cos\theta_1)}}{\sqrt{2g\ell(1 - \cos\theta_0)}}$$

$$\sqrt{2g\ell} (\sqrt{1 - \cos\theta_0} - \sqrt{1 - \cos\theta_1}) = V_m \quad \dots (i)$$

$$m\sqrt{2g\ell} (\sqrt{1 - \cos\theta_0} + \sqrt{1 - \cos\theta_1}) = MV_m \quad \dots (ii)$$

Dividing (ii) by (i) we get

$$\frac{(\sqrt{1 - \cos\theta_0} + \sqrt{1 - \cos\theta_1})}{(\sqrt{1 - \cos\theta_0} - \sqrt{1 - \cos\theta_1})} = \frac{M}{m}$$

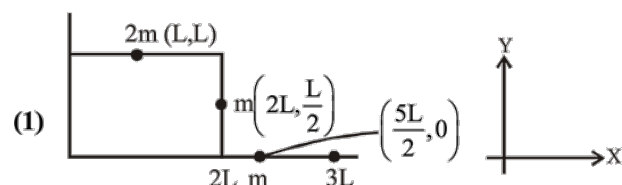
By componendo and dividendo rule

$$\frac{m - M}{m + M} = \frac{\sqrt{1 - \cos\theta_1}}{\sqrt{1 - \cos\theta_0}} = \frac{\sin\left(\frac{\theta_1}{2}\right)}{\sin\left(\frac{\theta_0}{2}\right)}$$

$$\Rightarrow \frac{M}{m} = \frac{\theta_0 - \theta_1}{\theta_0 + \theta_1} \Rightarrow M = m \frac{\theta_0 - \theta_1}{\theta_0 + \theta_1}$$

12 Jan Morning

Q6



x-coordinate of centre of mass is

$$X_{cm} = \frac{2mL + 2mL + \frac{5mL}{2}}{4m} = \frac{13}{8}L$$

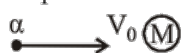
y-coordinate of centre of mass is

$$Y_{cm} = \frac{2m \times L + m \times \left(\frac{L}{2}\right) + m \times 0}{4m} = \frac{5L}{8}$$

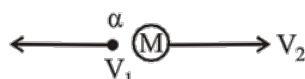
12 Jan Morning

Q7

- (4) Using conservation of momentum,
 $mv_0 = mv_2 - mv_1$



After collision



$$\frac{1}{2}mv_1^2 = 0.36 \times \frac{1}{2}mv_0^2$$

$$\Rightarrow v_1 = 0.6v_0$$

The collision is elastic. So,

$$\frac{1}{2}MV_2^2 = 0.64 \times \frac{1}{2}mv_0^2 \quad [\because M = \text{mass of nucleus}]$$

$$\Rightarrow V_2 = \sqrt{\frac{m}{M}} \times 0.8V_0$$

$$mV_0 = \sqrt{mM} \times 0.8V_0 - m \times 0.6V_0$$

$$\Rightarrow 1.6m = 0.8\sqrt{mM}$$

$$\Rightarrow 4m^2 = mM$$

$$\therefore M = 4m$$

12 Jan Evening

Q8

- (1) Acceleration of centre of mass (a_{cm}) is given by

$$\therefore \vec{a}_{cm} = \frac{m_1\vec{a}_1 + m_2\vec{a}_2 + \dots}{m_1 + m_2 + \dots}$$

$$= \frac{(2m)a\hat{j} + 3m \times a\hat{i} + ma(-\hat{i}) + 4m \times a(-\hat{j})}{2m + 3m + 4m + m}$$

$$= \frac{2a\hat{i} - 2a\hat{j}}{10} = \frac{a}{5}(\hat{i} - \hat{j})$$

8 April Morning

Q9

$$(2) \quad m_1v_1 + m_2v_2 = m_1v_2 + m_2v_1$$

$$\text{or } m_1v_1 + (0.5m_1)v_2 = m_1 \cdot 0.5v_1 + (0.5m_1)v_4$$

$$\text{On solving, } v_1 = v_4 - v_2$$

8 April Evening

Q10

- (4) With respect to point θ , the CM of the cut-off portion

$$\left(\frac{a}{4}, \frac{b}{4}\right)$$

$$\text{Using, } x_{CM} = \frac{MX - mx}{M - m}$$

$$= \frac{M \times 0 - \frac{M}{4} \times \frac{a}{4}}{M - \frac{M}{4}} = -\frac{a}{12}$$

$$\text{and } y_{CM} = -\frac{b}{12}$$

So CM coordinates are

$$x_0 = \frac{a}{2} - \frac{a}{12} = \frac{5a}{12}$$

$$\text{and } y_0 = \frac{b}{2} - \frac{b}{12} = \frac{5b}{12}$$

8 April Evening

Q11

- (4) For head on elastic collision we have

$$V_1 = \frac{(m_1 - m_2)u_1}{m_1 + m_2} + \frac{2m_2u_2}{m_1 + m_2}$$

$$\text{Here } m_1 = 2\text{kg, } u_1 = x, u_2 = 0,$$

$$v_1 = x/4$$

$$\therefore \frac{x}{4} = \frac{(2 - m_2)x}{2 + m_2} \Rightarrow m_2 = 1.2\text{kg}$$

9 April Morning

Q12

$$(2) \quad m(2v) + 2mv = 0 + 2mv' \cos 45^\circ$$

$$\text{or } v' = 2\sqrt{2}v$$

9 April Evening

Q13

$$(3) \quad mv = (m + M) V'$$

$$\text{or } v = \frac{mv}{m + M} = \frac{mv}{m + 4m} = \frac{v}{5}$$

Using conservation of ME, we have

$$\frac{1}{2}mv^2 = \frac{1}{2}(m + 4m)\left(\frac{v}{5}\right)^2 + mgh$$

$$\text{or } h = \frac{2}{5} \frac{v^2}{g}$$

9 April Evening

Q14

(1) Apply conservation of linear momentum in X and Y direction for the system then

$$M(10\cos 30^\circ) + 2M(5\cos 45^\circ) = 2M(v_1\cos 30^\circ) + M(v_2\cos 45^\circ)$$

$$5\sqrt{3} + 5\sqrt{2} = \sqrt{3}v_1 + \frac{v_2}{\sqrt{2}} \quad \dots(1)$$

Also

$$2M(5\sin 45^\circ) - M(10\sin 30^\circ) = 2Mv_1\sin 30^\circ - Mv_2\sin 45^\circ$$

$$5\sqrt{2} - 5 = v_1 - \frac{v_2}{\sqrt{2}} \quad \dots(2)$$

Solving equation (1) and (2)

$$(\sqrt{3} + 1)v_1 = 5\sqrt{3} + 10\sqrt{2} - 5 \Rightarrow v_1 = 6.5 \text{ m/s}$$

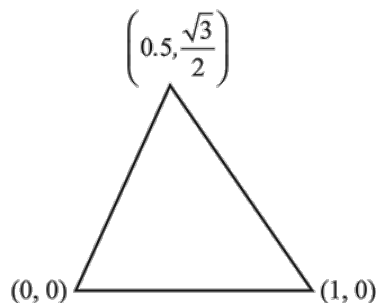
$$v_2 = 6.3 \text{ m/s}$$

10 April Morning

Q15

$$(3) \quad x_{\text{cm}} = \frac{50 \times 0 + 100 \times 1 + 150 \times 0.5}{50 + 100 + 150}$$

$$= \frac{7}{12} \text{ m}$$



$$y_{\text{cm}} = \frac{50 \times 0 + 100 \times 0 + 150 \times \frac{\sqrt{3}}{2}}{50 + 100 + 150}$$

$$= \frac{\sqrt{3}}{4} \text{ m}$$

12 April Evening